

Material Readiness Roadmap

Part 1

Integrated Provisioning Services
Airbus Services Solutions

TABLE OF CONTENTS

1. INTRODUCTION	2
2. INTRODUCTION AND MATERIAL READINESS PROCESS	2
3. PROVISIONING PROCESS TIMESCALE	2
4. THE MATERIAL READINESS SET-UP MEETING	3
4.1. DOCUMENTATION	4
4.2. AGENDA AND FORMAT OF THE MATERIAL READINESS CONFERENCE	5
5. PREPARING THE MATERIAL READINESS CONFERENCE	5
6. THE MATERIAL READINESS CONFERENCE	5
6.1. LINE REPLACEABLE UNITS	5
6.2. OTHER MATERIAL CATEGORIES	6
6.3. CONCLUSION AND REVIEW	6
7. POST MATERIAL READINESS CONFERENCE	6
8. PREVIOUSLY OWNED AIRCRAFT AND MATERIAL READINESS (MR) REVIEW	7
9. MATERIAL AND LOGISTICS ENGINEERING	8
10. ABBREVIATION LIST	9

1. INTRODUCTION

This introduction to the Material Readiness Roadmap process provides a guide to the activities that take place in order to ensure the successful provisioning of an Airbus customer. The Material Readiness (MR) is for unscheduled maintenance only and ensures flight operation of your aircraft.

It will illustrate the activities of the Integrated Provisioning Services department from the initial involvement in sales campaigns up to the support of previously owned and leased aircraft. In this brochure other documents will be referenced, which are available from or via the Integrated Provisioning Services department.



Tamas Stefka
Senior Director Integrated Provisioning Services

2. INTRODUCTION AND MATERIAL READINESS PROCESS

The Material Readiness and its process are to ensure revenue flight dispatch to the Airbus customers.

As defined in ATA SPEC 2000, the MR process is concerned with supporting a new aircraft fleet or an end item for an initial period of operation. Therefore the Integrated Provisioning Services team deals as a service provider for the aircraft maintenance by

- providing aircraft spares data for planning and purchasing purposes
- planning and recommendation of spares required for the Line and Base Maintenance support of Airbus aircraft
- consulting and training in material management processes

3. PROVISIONING PROCESS TIMESCALE

The traditional timescale of the MR process, as defined in the Supplier Support Conditions (SSC), has emerged over the years from the broad outline specified in the relevant chapters of both ATA SPEC 2000 (Chapter 1) and the World Airlines Suppliers Guide WASG (Chapter 3), which was replaced by the Common Support Data Dictionary CSDD.

According to ATA SPEC 2000 the transmission of provisioning data: "...starts well in advance of the first delivery to permit ordering and establishment of support inventories in time for initial operations."

The industry standards that have developed are illustrated in *Figure 1*.

According to the Airbus SSC, the MR process should begin: "...-24 months before delivery of the first aircraft...for long lead time items. " and "...- 18 months before delivery of the first aircraft ...for S File items."

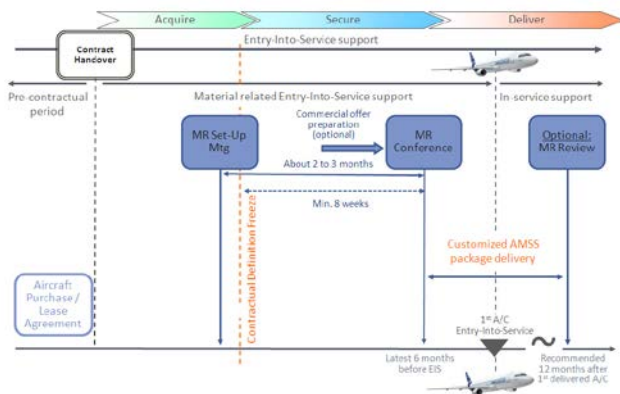


Figure 1: Material Readiness Roadmap

However, the timeframe given in the SSC does not comply with modern day demands and Airbus has geared itself to a more flexible schedule in order to provide shorter lead times for entry into service.

Airbus currently offers an aircraft production lead-time of seventh months for Single Aisle aircraft (A318 / A319/ A320/ A321) without long lead-time items. This has affected the time scale for the Material Readiness Set-up Meeting, which has been reduced by approximately 15 months.

4. THE MATERIAL READINESS SET-UP MEETING

The Material Readiness (MR) Set-up Meeting, unlike the Material Readiness (MR) Conference, is an obligatory aspect of the MR process as stated in ATA SPEC 2000 Chapter 1: "...a pre-provisioning conference will be held between the airframe/engine manufacturer or BFE supplier to plan the provisioning programme prior to the release of the first data."

Usually representatives of the customer will visit Airbus Material, Logistics and Suppliers in Hamburg for the MR Set-up Meeting. The primary objectives of the MR Set-up Meeting are:

- to confirm criteria concerning the aircraft and the airline's maintenance ability
- to ascertain the customer's proposed operating parameters
- to decide which data and documents the customer desires
- to prepare the agenda and format of the Material Readiness Conference
- to ascertain the customers specific requirements for the Material Readiness Conference
- to consider which personnel (e.g. Material Controllers, Supply Managers, Purchasers) should attend the conference
- to mutually set up the strategy for the provisioning exercise and future maintenance

The first three objectives are necessary in order to begin the process of producing spares recommendation documentation. It is necessary to confirm at least the initial MSN(s) in order to begin allocating customised data.

It is also essential to establish other basic criteria in respect of the customer's maintenance capability, for example whether the customer intends purchasing an Automatic Test Equipment (ATE),

as this will have an impact on the turnaround time for appropriate repairable items.

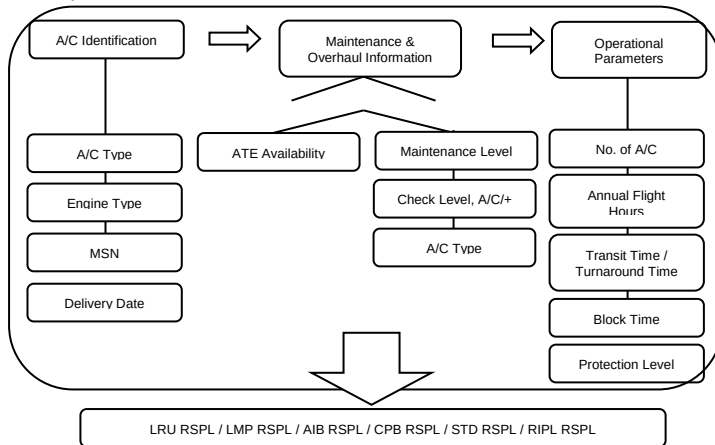


Figure 2: Information process for Provisioning Document publication.

Operational parameter, which are determined by the customer are the essential variables to calculate the spare recommendation. The principles to the mathematical model are subject of the accompanying handbook, *"Integrated Provisioning Services, Part 3 - The Mathematical Model"*.

A Provisioning Training can be provided in conjunction with the MR Set-up Meeting. It entails a familiarization with the provisioning documents and methodology, Airbus supplier management, spares data management etc.

4.1. DOCUMENTATION

At the Material Readiness Conference the Airline receives the Initial Issue of the Provisioning Documentation. This includes both Recommended Spares Parts List (RSPL) and the Detailed Parts Lists (DPL). Those documents are provided in excel format.

Standard Provisioning Documents:

RSPL:

- Line Replaceable Units (LRU)
- Line Maintenance Breakdown Parts of LRUs (LMP)
- Cockpit Push Buttons (CPB)
- Standard Hardware (STD)

DPL:

- Airbus Proprietary Parts (AIB)

Supplementary Provisioning Documents:

- Airbus Material Maintenance Forecast (AMMF)
- Aircraft Component Replacement Data (ACRD)
- Material Storage Data (MSD)

The RSPLs can be supplied in several sorts, including part number sort, manufacturer sort, end item sort (for LMPs) and ATA 100 Chapter sort depending on the customer's preferences. The RSPLs are supplied in Excel format. The format, content and further descriptions of the RSPLs are described in detail in the accompanying booklet, *"Integrated Provisioning Services, Part 2 – The Provisioning Documentation."*

4.2. AGENDA AND FORMAT OF THE MATERIAL READINESS CONFERENCE

At the stage of the MR Set-up of the provisioning process it must also be decided whether to invite suppliers or not. If suppliers should be invited, the customer may choose which suppliers they wish to see. If no suppliers are invited, all the provisioning data will be presented by Airbus.

Finally, the date and attendees for the MR conference must be agreed. This event usually takes place at Airbus Material, Logistics and Suppliers in Hamburg.

5. PREPARING THE MATERIAL READINESS CONFERENCE

If a MR Conference is mutually agreed, the MR Set-up Meeting is used to prepare the background for the conference.

Following the MR Set-up Meeting, the process of producing, revising, and refining the initial provisioning documentation will begin. An initial set of the RSPLs will be produced and copies of the relevant pages will be sent to the invited suppliers for their comments in order to update the information. The process of revising data such as part number, effectivity and prices will continue throughout the Material Readiness process to reflect the latest updates. In order to minimize the operational threat of such updates, Airbus has configuration guarantees in the Supplier Support Conditions (SSC).

6. THE MATERIAL READINESS CONFERENCE

6.1. LINE REPLACEABLE UNITS

The primary focus of the MR Conference will be supplier LRUs. The reason for this is that they represent an overwhelming percentage of the investment for IP whilst representing only a fraction of the quantity of part numbers as illustrated in *Figure 3*. Other topics discussed during the MR Conference are internal Airbus topics concerning GSE & Tools, LMPs, AIBs, STDs & PUBs, financial and invoicing procedures, purchase order administration procedures and a conclusion and review session.

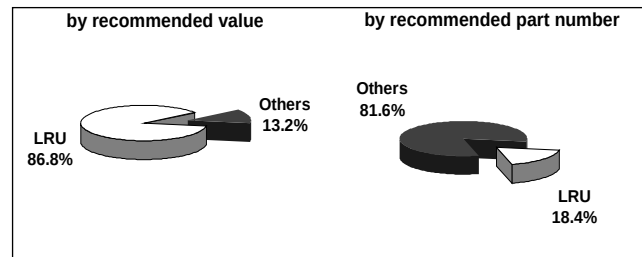


Figure 3: Comparison of material categories - % share of LRUs in an IP package by value and quantity

The time dedicated to LRUs during the conference is therefore related to their relative importance. Typically, a conference will last one week. Suppliers, who represent the majority of the proposed LRU investment, will be invited to the conference on behalf of the customers.

Their presentations allow them to:

- Briefly explain and illustrate their company's background and product support including their AOG services, repair procedures and spares finance policies
- Present their product
- Answer questions raised by the customer
- Allow to enter technical discussions

The suppliers will also provide information concerning their company with appropriate contact details to hand over to the customer. The quality of the presentation and the documents provided are closely monitored by the provisioning department and any shortfall from the expected levels of performance are fed back to the supplier's management.

The LRU RSPL for spares from suppliers with smaller values of investment will be presented by Airbus. Usually the presentation will go through the LRUs, supplier by supplier, covering all the spares recommended. Thus the total amount of time spent on LRUs may take seventy percent of the time of a MR Conference. The other material categories and topics of the conference cover the remaining time.

6.2. OTHER MATERIAL CATEGORIES

These categories are dealt with in a different manner to LRUs. The aim of these sessions is to explain the layout, the logic, and how the RSPLs of these categories interface with the LRU RSPL and between themselves. Therefore these sessions concentrate on explaining the documents rather than recommending and

explaining individual items. This allows the customer to analyse the documents and make their purchase decision later on.

The responsible departments within Airbus hold the sessions concerning purchase order administration, financing and invoicing. These sessions are held in order to inform the airline about the current procedures for those tasks.

6.3. CONCLUSION AND REVIEW

This session will reiterate the action that needs to be taken following the Material Readiness conference. This typically consists of outstanding questions from the conference and the actions that must be taken to answer them, confirming which documents to be sent to which customer's email-address and what the revision cycle of the documents will be during the remaining time of the provisioning process.

The revision cycle will depend on the RSPL. The W File, PICL, Airbus Proprietary Parts, Standard Hardware, and Cockpit Push Button RSPLs usually are revised on request. The LRU and LMP documents, which are subject to a more significant change during the provisioning process, are revised as mutually agreed. This is usually every three months, in line with ATA SPEC 2000: "Revisions to the provisioning data, ... will be submitted to the customer on a revision cycle as negotiated following the original submission." and the Common Support Data Dictionary (CSDD): "...revisions will be issued every 60 to 90 days."

7. POST MATERIAL READINESS CONFERENCE

The aim of the MR conference is to enable the customer to make their purchase decision. Following the conference, the provisioning process mainly consists of Airbus revising and refining the

provisioning information, as entry into service approaches, with the customer purchasing the equipment they consider to be necessary.

The target for the airline is to have at least 50% of their provisioning stock delivered at least three months in advance of, and a complete stock three months after, entry into service. This is fixed in Exhibit “H” (2.7.1.1 and 2.7.1.4) of a contract where a supplier must supply:

“...at least fifty percent (50%) of the ordered quantity.....three (3) months before delivery...” and “..one hundred percent (100%) of the ordered quantity of each item,.....three (3) months after delivery of the last aircraft.”

As stated previously the length of this period of provisioning data cycles and purchasing may be extended over a period of years, depending on the delivery schedule of the aircraft.

The Material Readiness process contractually ends three months after the delivery of the last aircraft of a specific order. However this does not entail the end of the provisioning support – as option, customers can order data and services as chargeable service in accordance with the Airbus Customer Services Catalogue.

8. PREVIOUSLY OWNED AIRCRAFT AND MATERIAL READINESS (MR) REVIEW

The provisioning period for previously owned aircraft and MR Review may be much shorter than for a customer purchasing new build aircraft. If an airline acquires additional aircraft second hand and/or through secondary lease agreements, they will require the

information for provisioning immediately as the lead-time for entry into service will be shortened and constrains like the Configuration Definition Freeze date do not apply.

Under those circumstances, the functions of a normal provisioning exercise usually cannot be carried out. Therefore for previously owned aircraft the usual provisioning process consists of extracting and customising the RSPLs and handing over the documentation to the customer. The configuration data is equivalent to the configuration shown in the Illustrated Parts Catalogue valid at that point of time.

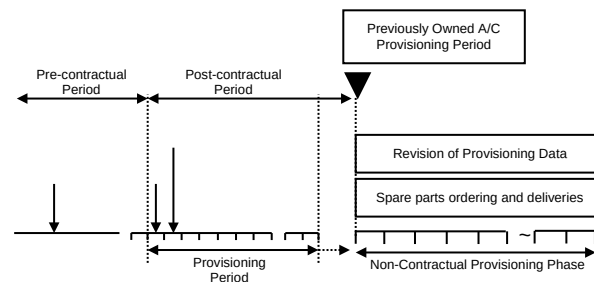


Figure 4: Previously Owned A/C lead-time for provisioning.

A MR Review is offered on customer request for an established fleet and inventories when airline route structure and/or fleet configuration are altered, main base and station inventories are assessed for surplus or new requirements. Also MR Reviews are conducted for new build aircraft if delivery streams are spread over a period of several years. In that case, the MR Review exercise

may be similar to that of the MR process. This will depend on the customer's feeling to run through the whole process again.

9. MATERIAL AND LOGISTICS ENGINEERING

The main task of the Material and Logistics Engineering team is the technical analyses of spares in order to reduce the warehouse stock as an important instrument of cost saving. To achieve this aim we are applying the following approach:

- Increasing interchangeability
- Harmonization and standardization of parts taking into account technical needs
- Design contribution in new and running A/C programs regarding to spares
- Essentiality and in-service MTBUR data process management

10. ABBREVIATION LIST

A/C	Aircraft	SIF	Sales Investment Forecast
AIB	Airbus Proprietary Parts	STD	Standard Hardware Materia
AMMF	Airbus Material Maintenance Forecast	SSC	Supplier Support Conditions
AOG	Aircraft on Ground	WASG	World Airlines Suppliers Guide
ATA	Air Transport Association		
ATE	Automatic Test Equipment		
BFE	Buyer Furnished Equipment		
CML	Consumable Material List		
CPB	Cockpit Push Buttons		
CSDD	Common Support Data Dictionary		
GSE	Ground Support Equipment		
IPC	Illustrated Parts Catalogue		
LMP	Line Maintenance Breakdown Part of LRU		
LRU	Line Replaceable Unit		
MR	Material Readiness		
MSD	Material Storage Data		
MSN	Manufacturer's Serial Number		
MTBUR	Mean Time Between Unscheduled Removals		
PICL	Pool Item Candidate List		
Adv.RPD	Advanced Resource Planning Data		
RSPL	Recommended Spare Parts List		

© AIRBUS S.A.S. All rights reserved. Confidential and proprietary document.

This document and all information contained herein is the sole property of AIRBUS S.A.S. No intellectual property rights are granted by the delivery of this document or the disclosure of its content. This document shall not be reproduced or disclosed to a third party without the express written consent of AIRBUS S.A.S.

This document and its content shall not be used for any purpose other than that for which it is supplied. The statements made herein do not constitute an offer. They are based on the mentioned assumptions and are expressed in good faith. Where the supporting grounds for these statements are not shown, AIRBUS S.A.S. will be pleased to explain the basis thereof.

AIRBUS, its logo, A300, A310, A318, A319, A320, A321, A330, A340, A350, A380, A400M are registered trademarks



Airbus Services Solutions
Integrated Provisioning Services
Weg beim Jäger 150
22335 Hamburg, Germany
email: material-readiness.provisioning@airbus.com
Phone: + 49 (40) 50 76 – 0